

Exact Parameter Identification in PET Pharmacokinetic Modeling Using the Irreversible Two Tissue Compartment Model

CSE 402 Numerical Analysis, Simulation and Modeling — Section A1, Group 5

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Abstract

Dynamic positron emission tomography (PET) estimates regional kinetic parameters from tissue time–activity curves, but conventional quantification also requires an arterial input function obtained through invasive blood sampling. We reproduce the numerical core of Holler, Morina, and Schramm’s exact-identifiability analysis for the irreversible two-tissue compartment model using hand-written numerical methods. The implementation includes LU factorization with partial pivoting, Householder QR least squares, nonuniform Simpson integration, a seeded linear congruential random-number generator with Box–Muller and Poisson samplers, power and inverse-power iterations, an analytic 104-by-23 Jacobian, and an iteratively regularized Gauss–Newton method (IRGNM). The closed-form tissue model agrees with an independent Radau ODE solution to 1.73×10^{-13} , and the quadrature path agrees to 3.29×10^{-10} . In 65 accepted noiseless tissue-only fits, the four ratios $K_{1,\text{est}}^i/K_{1,\text{true}}^i$ coincide to a worst spread of 1.83×10^{-8} although individual K_1 error reaches 127%; adding one early plasma measurement reduces $|\zeta - 1|$ to at most 1.73×10^{-7} . Correcting a stopping-rule scaling error and adopting the paper’s per-setup settings cut Monte Carlo blow-ups from 494 to 43 and match the paper’s failure counts at high and normal count. With noise, the signature remains visible at high and normal count but fails at low count. Regularization reduces fitted-parameter variance by factors of 1.76 and 4.39. These results reproduce the paper’s structural conclusion while exposing limits caused by TAC-level noise, initialization, stopping conventions, and floating-point evaluation.

Code: github.com/Undying2021Dreams/Numerical-Sessional-Project

CCS Concepts

• **Mathematics of computing** → **Numerical analysis**; *Probabilistic algorithms*; *Numerical optimization*.

Keywords

dynamic PET, pharmacokinetic modeling, identifiability, inverse problems, IRGNM, regularization, numerical analysis, Monte Carlo simulation

1 Introduction and Base Paper Review

Dynamic PET records tracer concentration over time and uses a compartment model to infer physiological rates. For an irreversible tracer, the regional parameters K_1 , k_2 , and k_3 respectively describe transport from plasma into tissue, return from tissue to plasma, and irreversible trapping. Clinical estimation normally depends on the metabolite-corrected arterial plasma concentration $C_P(t)$. Directly measuring that curve requires repeated blood draws and chemical processing; image-derived measurements more naturally provide whole-blood concentration $C_{WB}(t)$.

Holler, Morina, and Schramm [6] analyzed simultaneous fitting of several regions under the irreversible two-tissue compartment model. Their main result states that, subject to technical assumptions and sufficiently many time samples, tissue observations identify every k_2^i and k_3^i and identify all K_1^i up to one common scale. Blood information removes that scale ambiguity. They connect the noiseless theorem to noisy observations through Tikhonov regularization and an iteratively regularized Gauss–Newton method (IRGNM).

Our objective is not to reproduce the paper’s digits. Its full simulation includes a sinogram and OSEM reconstruction chain, whereas ours adds calibrated noise directly to regional time–activity curves (TACs). Agreement is therefore defined *before inspecting the results* as reproduction of mathematical invariants and qualitative trends: the common K_1 scale in tissue-only fitting, removal of that ambiguity by arterial information, decreasing error with decreasing noise, and reduced variance under regularization. Absolute failure counts are reported but are not expected to match.

The main contributions are: (i) a from-scratch numerical stack with no library solvers in the production modules, each component benchmarked against an independent reference; (ii) three independent evaluations of the forward model; (iii) an analytic Jacobian and regularized nonlinear solver; (iv) a 960-cell Monte Carlo study; (v) a direct parameter-level experiment for the paper’s identifiability theorem; (vi) a record of the numerical enhancements we tried, including the ones that did not reach the final pipeline; and (vii) a controlled before-and-after evaluation of each improvement (Section 4.5). The public implementation and all regeneration scripts are

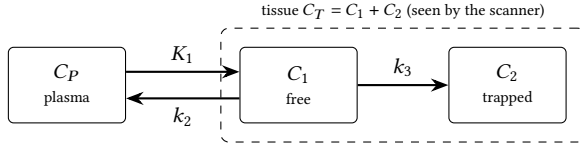


Figure 1: Irreversible two-tissue compartment model for one region. Tracer enters tissue at rate K_1 , returns to plasma at k_2 , and is trapped at k_3 ; trapped tracer never leaves.

available at <https://github.com/Undying2021Dreams/Numerical-Sessional-Project>.

2 Methodology

2.1 Irreversible two-tissue model

Figure 1 shows the model. For each region, the exchangeable and trapped tissue compartments satisfy

$$\dot{C}_1(t) = K_1 C_P(t) - (k_2 + k_3) C_1(t), \quad (1)$$

$$\dot{C}_2(t) = k_3 C_1(t), \quad C_T(t) = C_1(t) + C_2(t). \quad (2)$$

Writing $a = k_2 + k_3$, the tissue concentration is evaluated as

$$C_T(t) = K_1 \frac{k_2}{a} \int_0^t e^{-a(t-s)} C_P(s) ds + K_1 \frac{k_3}{a} \int_0^t C_P(s) ds. \quad (3)$$

The first term is a fading memory of recent input (the free compartment); the second is a running total of everything ever delivered, scaled by the trapping fraction k_3/a . The shared arterial input is a degree-four polyexponential,

$$C_P(t) = \sum_{j=1}^4 \lambda_j e^{\mu_j t}, \quad (4)$$

and the parent-plasma fraction is $f(t) = A e^{\xi_1 t} + (1-A) e^{\xi_2 t}$, giving $C_{WB} = C_P/f$. The measured PET activity is $C_{PET} = (1 - V_B) C_T + V_B C_{WB}$ with $V_B = 0.05$.

The unknown vector contains 23 parameters: four λ_j , four μ_j , three parameters of f , and (K_1, k_2, k_3) for each of four brain regions. The 11 blood parameters are shared by all regions, while each region owns its three kinetic rates. Ground-truth constants and the 25-frame schedule are taken from the paper’s Section 5.1 and Table 2 and stored once in `src/config.py`.

2.2 Software architecture

Figure 2 shows how the hand-written primitives feed the model and the solver. Each box is a module under `src/`; every arrow is a real call dependency. No higher layer calls a library routine: when it needs a linear solve, an integral, an eigenvalue, or a random number, it calls the primitive layer.

2.3 Hand-written numerical primitives

All production algorithms under `src/` are implemented without SciPy solvers, NumPy linear solves, or NumPy random sampling; NumPy is used only for array storage and elementwise arithmetic. An automated test walks the syntax tree of every module under `src/`, resolves import aliases, and fails on any banned symbol. This subsection describes each primitive in the form it is implemented.

Algorithm 1 LU with partial pivoting (`lu_factor`, `lu_solve`)

Require: $A \in \mathbb{R}^{n \times n}$, $b \in \mathbb{R}^n$

```

1: for  $k = 1$  to  $n - 1$  do
2:    $p \leftarrow \arg \max_{i \geq k} |a_{ik}|$ ; swap rows  $k$  and  $p$  of  $A$ ; record  $p$ 
3:   if  $|a_{kk}| < 10^{-300}$  then
4:     raise singular matrix
5:   end if
6:   for  $i = k + 1$  to  $n$  do
7:      $\ell_{ik} \leftarrow a_{ik}/a_{kk}$   $\triangleright |\ell_{ik}| \leq 1$ 
8:      $a_{i,k+1:n} \leftarrow a_{i,k+1:n} - \ell_{ik} a_{k,k+1:n}$ 
9:   end for
10: end for
11: apply the recorded swaps to  $b$ 
12: solve  $Ly = b$  by forward substitution
13: solve  $Ux = y$  by back substitution
14: return  $x$ 
```

Algorithm 2 Householder least squares (`qr_factor`, `qr_lstsq`)

Require: $A \in \mathbb{R}^{m \times n}$ with $m \geq n$, $b \in \mathbb{R}^m$

```

1:  $R \leftarrow A$ 
2: for  $k = 1$  to  $n$  do
3:    $x \leftarrow R_{k:m,k}$ ; if  $\|x\| = 0$  then store  $v_k = 0$  and continue
4:    $\alpha \leftarrow -\text{sign}(x_1) \|x\|_2$ ;  $v \leftarrow x - \alpha e_1$ ;  $v_k \leftarrow v/\|v\|$ 
5:    $R_{k:m,k:n} \leftarrow R_{k:m,k:n} - 2v_k (v_k^T R_{k:m,k:n})$ 
6: end for
7:  $c \leftarrow b$ ; for  $k = 1$  to  $n$ :  $c_{k:m} \leftarrow c_{k:m} - 2v_k (v_k^T c_{k:m})$   $\triangleright c = Q^T b$ 
8: solve  $R_{1:n,1:n} x = c_{1:n}$  by back substitution
9: return  $x$  and residual norm  $\|c_{n+1:m}\|$ 
```

2.3.1 LU factorization with partial pivoting. Linear systems $Ax = b$ are solved by Doolittle elimination with row pivoting, $PA = LU$, where L is unit lower triangular and U is upper triangular [2, 3]. At step k the row with the largest $|a_{ik}|$, $i \geq k$, is swapped into the pivot position, so every multiplier satisfies $|\ell_{ik}| \leq 1$ and no division by a small pivot can amplify rounding error. A pivot below 10^{-300} in magnitude raises an explicit singular-matrix error instead of returning garbage. The solve is then two triangular sweeps, $Ly = Pb$ forward and $Ux = y$ backward (Algorithm 1). The cost is $\frac{2}{3}n^3$ flops for the factorization and $2n^2$ per right-hand side. LU is used inside inverse power iteration and in the normal-equations path of IRGNM that we keep for comparison.

2.3.2 Householder QR and linear least squares. For an $m \times n$ matrix with $m \geq n$, column k is reduced by the reflector $H_k = I - 2v_k v_k^T$ with

$$v_k = \frac{x - \alpha e_1}{\|x - \alpha e_1\|}, \quad \alpha = -\text{sign}(x_1) \|x\|_2, \quad (5)$$

where x is the part of column k on and below the diagonal (Algorithm 2). Choosing the sign of α opposite to x_1 prevents cancellation in $x_1 - \alpha$. The matrix $Q = H_1 \cdots H_n$ is never formed: only the vectors v_k are stored, and $Q^T b$ is computed by re-applying them. For least squares, writing $Q^T b = [c_1; c_2]$ gives the solution of $R_{1:n,1:n} x = c_1$ and the exact residual norm $\|c_2\|$. QR costs about twice as many flops as LU on a square system, but it works on A directly instead of $A^T A$, so the condition number is not squared. This is why QR is the production solver inside IRGNM, and it is also the regression tool behind every log–log slope fit and the Patlak line in this report.

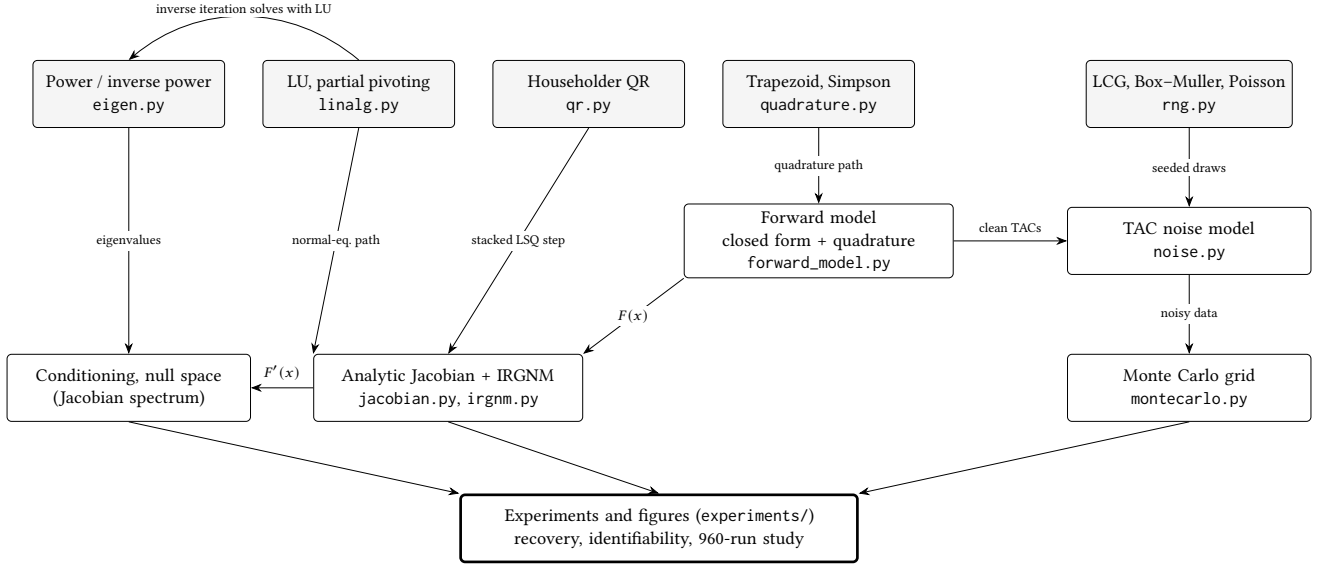


Figure 2: Module dependency graph. Shaded boxes are the hand-written primitives (Track A). The model, solver and experiment layers only reach numerical functionality through them; NumPy and SciPy appear only in tests and marked benchmarks (Track B).

2.3.3 Trapezoid and Simpson integration on nonuniform grids. The composite trapezoid rule is $\sum_i \frac{1}{2}(x_{i+1} - x_i)(y_i + y_{i+1})$, with error $O(h^2)$. Simpson's rule must also work on the nonuniform grids used by the forward model, so it is implemented by integrating the quadratic interpolant through each consecutive node triple $x_0 < x_1 < x_2$ exactly. With $h_0 = x_1 - x_0$, $h_1 = x_2 - x_1$ and $w = h_0 + h_1$,

$$\int_{x_0}^{x_2} p_2 dx = \frac{w}{6} \left[\left(2 - \frac{h_1}{h_0}\right) y_0 + \frac{w^2}{h_0 h_1} y_1 + \left(2 - \frac{h_0}{h_1}\right) y_2 \right], \quad (6)$$

which reduces to the textbook weights $\frac{h}{3}(1, 4, 1)$ when $h_0 = h_1 = h$. If the number of intervals is odd, the last interval is closed with one trapezoid step and the fallback is reported. On a uniform grid the leading error of the composite rule is

$$E(h) \approx \frac{h^4}{180} [f'''(b) - f'''(a)] + O(h^6), \quad (7)$$

so halving h divides the error by 16. When $f'''(a) = f'''(b)$ the h^4 term cancels and the observed order becomes 6; Section 4.1.2 shows this happening.

2.3.4 Random numbers: LCG, Box-Muller and Poisson. Uniform numbers come from a 64-bit linear congruential generator [9]

$$s_{k+1} = (a s_k + c) \bmod 2^{64}, \quad u_k = \lfloor s_{k+1} / 2^{11} \rfloor \cdot 2^{-53}, \quad (8)$$

with Knuth's MMIX constants $a = 6364136223846793005$ and $c = 1442695040888963407$. Only the top 53 bits are used, because the low bits of an LCG have short periods, and 53 bits fill a float64 mantissa exactly. The integer seed is first passed through a SplitMix64 avalanche so that nearby seeds do not start in related states, and sub-seeds for each experiment are derived deterministically from a SHA-256 hash of the root seed and string labels. Normal variates use the Box-Muller transform [1],

$$z_1 = \sqrt{-2 \ln u_1} \cos(2\pi u_2), \quad z_2 = \sqrt{-2 \ln u_1} \sin(2\pi u_2), \quad (9)$$

with u_1 drawn from the open interval $(0, 1)$ so the logarithm stays finite. Poisson variates use Knuth's multiplication algorithm, which multiplies uniforms until the product drops below $e^{-\lambda}$, for $\lambda \leq 30$, and the rounded normal approximation $\max(0, \text{round}(\lambda + \sqrt{\lambda} z))$ above it. The crossover matters: Knuth's method costs $O(\lambda)$ uniforms per draw, and for $\lambda \gtrsim 745$ the threshold $e^{-\lambda}$ underflows to zero, so the loop would never terminate.

2.3.5 Power and inverse power iteration. The largest eigenvalue of the symmetric matrix $M = F'^T F'$ is found by the power method with a Rayleigh-quotient stopping test (relative change below 10^{-13}), and the full spectrum by repeated Hotelling deflation $M \leftarrow M - \lambda v v^T$. The smallest eigenvalue, which sets the conditioning, comes from inverse power iteration, where each step solves $M w = v$ with our LU factorization.

2.4 Stable forward evaluation and quadrature

Equation (3) has both a closed-form and an independently integrated implementation. Substituting the polyexponential (4) turns both integrals into sums of terms of the form $\int_0^t e^{cs} ds = t \phi_1(ct)$, with

$$\phi_1(z) = \frac{e^z - 1}{z}, \quad \phi_1(0) = 1. \quad (10)$$

Evaluated literally, ϕ_1 subtracts two nearly equal numbers when $|z|$ is small, and at $z = 0$ it is 0/0. We evaluate it as $\text{expm1}(z)/z$, which computes $e^z - 1$ without forming e^z first. For one exponential component, the convolution kernel is evaluated as

$$E(t, a, \mu) = t e^{\max(\mu, -a)t} \phi_1(-|a + \mu|t), \quad (11)$$

which avoids an artificial $0 \cdot \infty$ intermediate and stays finite when $a + \mu \rightarrow 0$. The Jacobian needs $\phi_1'(x) = 1 + (x - 1)\phi_2(x)$ with $\phi_2(x) = (e^x - 1 - x)/x^2$. Here expm1 alone is not enough, because the numerator still subtracts x from a nearly equal quantity; below

$|x| = 10^{-4}$, a threshold we measured rather than assumed, we switch to the Taylor series $\phi_2(x) = \frac{1}{2} + \frac{x}{6} + \frac{x^2}{24} + \dots$, whose terms are all added.

The independent path applies our nonuniform composite Simpson rule on a graded grid $t_i = t_{\max} u_i^3$ with u_i uniform on $[0, 1]$ and 1601 points. Grading places most nodes in the first minutes, where the injected bolus makes C_P change fastest. Three-point Lagrange weights are computed in local coordinates, as in (6). This change preserves Simpson's rule while avoiding cancellation from subtracting cubic antiderivatives at large absolute coordinates.

2.5 Analytic Jacobian and IRGNM

The forward operator $F : \mathbb{R}^{23} \rightarrow \mathbb{R}^{104}$ contains 100 regional TAC observations (4 regions times 25 frames) and four blood constraints $C_{WB}(s)f(s) - C_P(s)$ at the sampling times $s \in \{0.29, 2.25, 15, 57.5\}$ min. Its analytic Jacobian is derived from the closed form and verified columnwise against central differences. The Jacobian is block structured: each region's rows depend on the shared blood parameters and on that region's own three rates only, the tissue rows do not depend on the plasma-fraction parameters, and the blood rows do not depend on any tissue rate. The regularization and globalization choices follow established inverse-problem and nonlinear-optimization principles [4, 10]. At iteration i , the IRGNM step solves

$$\min_{\Delta x} \left\| \begin{bmatrix} F'(x_i) \\ \Lambda_i^{1/2} \end{bmatrix} \Delta x - \begin{bmatrix} y^\delta - F(x_i) \\ \Lambda_i^{1/2}(x_0 - x_i) \end{bmatrix} \right\|_2^2, \quad (12)$$

using Householder QR. Six blockwise regularization schedules follow ae^{-bi} : metabolic, arterial, and plasma-fraction blocks each have separate magnitudes, because the 23 parameters span a scale ratio of 2690. The adopted values are $a_\alpha = 4000$, $a_\beta = 100$, $a_\gamma = 200$, $b = \ln(2)/7$, and discrepancy factor $\tau = 6.8$. Each full step is projected onto the feasible domain. A normal-equations/LU path is retained for comparison; the production solver takes full steps without a line search. Figure 3 summarizes one run.

For noisy data, Morozov stopping compares the residual 2-norm with τ times the noise 2-norm. Since calibration returns an RMS discrepancy δ_y , experiments pass $\sqrt{n_{\text{obs}}} \delta_y$ where dimensionally required. Iteration-zero stops are recorded as trivial stops, not successful fits.

2.6 Identifiability statistic

With tissue data only, the transformation

$$K_1^i \mapsto \zeta K_1^i, \quad \lambda_j \mapsto \lambda_j / \zeta \quad (13)$$

leaves every regional TAC unchanged. We measure

$$s_\zeta = \frac{\max_i (K_{1,\text{est}}^i / K_{1,\text{true}}^i)}{\min_i (K_{1,\text{est}}^i / K_{1,\text{true}}^i)} - 1. \quad (14)$$

A small s_ζ together with a large $|\zeta - 1|$ is the predicted signature. We then restore a single C_P observation and measure the collapse of $|\zeta - 1|$.

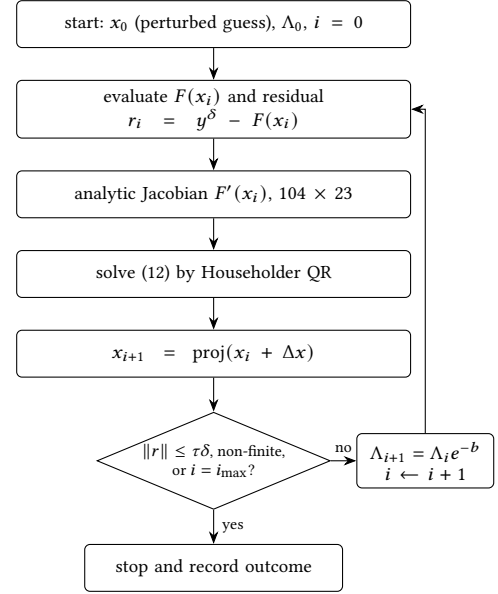


Figure 3: One IRGNM run. The regularization weights shrink geometrically, so early steps are cautious and later steps approach plain Gauss–Newton. Divergence (non-finite iterates) is recorded as a failure, never silently retried.

3 Implementation and Experiments

3.1 Synthetic data and measurement setups

The four regions are frontal, temporal, occipital, and white matter. Their ground-truth (K_1, k_2, k_3) values, taken from the base paper [6], are the control-group values of Jagust et al. [8]: $(0.157, 0.174, 0.118)$, $(0.161, 0.179, 0.096)$, $(0.177, 0.159, 0.088)$, and $(0.100, 0.161, 0.047)$. The acquisition contains 25 frames: 4×5 s, 4×10 s, 4×30 s, 2×60 s, 3×150 s, 6×300 s, and 2×600 s. Figure 4 shows the resulting curves.

Poisson-derived TAC noise and Gaussian blood noise were calibrated by bisection to target RMS discrepancies 0.003, 0.011, and 0.070 for high, normal, and low count. The experiment grid contains three setups, four noise levels (including noiseless), four initial perturbation levels $\delta_x \in \{0.1, 0.2, 0.3, 0.4\}$, and 20 deterministic seeds: 960 fits. Setup A fixes the plasma-fraction parameters and omits the blood block; Setup B fits all 23 parameters with clean C_{WB} ; Setup C fits all parameters with noisy C_{WB} . This repository's Setup A is therefore the deliberately tissue-only identifiability configuration and is not numerically identical to every setting in the paper's Table 1.

3.2 Benchmarks for the numerical building blocks

Every primitive is tested against a truth that is known independently of the code under test (Table 1). For linear systems we use manufactured solutions: we choose x , form $b = Ax$, and compare the computed x with the chosen one, with NumPy as a second referee. The Hilbert matrices $H_{ij} = 1/(i + j + 1)$ with $x = (1, \dots, 1)$ are a deliberately ill-conditioned stress test. Integration is checked

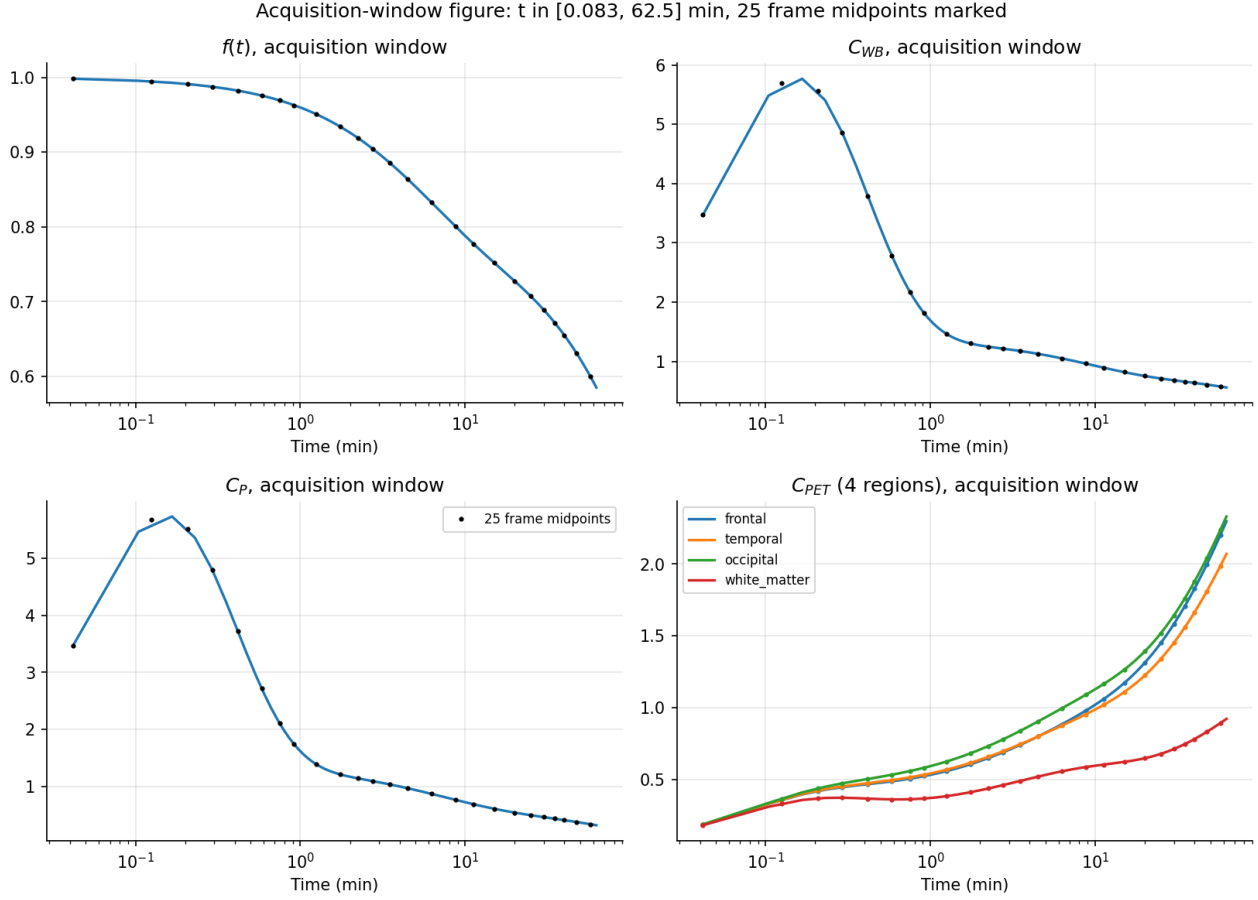


Figure 4: Synthetic plasma fraction, whole-blood input, plasma input, and four regional PET TACs over the 25-frame acquisition window. Dots mark frame midpoints.

on integrals with known values, the random generators against the moments and distribution functions of their target laws, and the forward model against two independent code paths. Convergence orders are measured as least-squares slopes on log-log data, computed with our own QR routine.

3.3 Verification protocol

Track A denotes the hand-written implementation; Track B denotes NumPy [5] and SciPy [13] used only as independent benchmarks in tests/ or experiments/. The forward model was checked against direct Simpson integration and `solve_ivp` with the implicit Radau method. The Jacobian was checked against central differences. Figures are generated with Matplotlib [7], and the experiment harness runs on Python [12]. Every stochastic run uses a derived seed, and generated JSON files record the root seed, configuration hash, source hash, Python/NumPy versions, and platform. The primary numerical artifacts use configuration hash 2f38ba062972 and source-hash prefix 9d11616d97b7. The latest one-command manifest records 18 of 18 scripts completing successfully in 2264.7 s on Windows with Python 3.10.6 and NumPy 2.2.6. A fresh full

test run reports 176 passed; the solver-ban guard remains part of that suite.

4 Results and Discussion

4.1 Numerical verification

Table 2 shows that the numerical building blocks reach the accuracy needed by the inverse problem. The analytic closed form is effectively indistinguishable from the independent stiff ODE reference. The graded-grid quadrature is less accurate but remains below 3.30×10^{-10} relative difference across all four regions and 25 frames. The following subsections give the detail behind each row.

4.1.1 Linear solvers. On 200 random systems of size 5 to 100, the mean relative residual $\|Ax - b\|/\|b\|$ is 2.67×10^{-16} for LU and 5.51×10^{-16} for QR, with maxima 4.65×10^{-16} and 9.22×10^{-16} (Figure 5). Both are at the level of the unit round-off $\approx 1.1 \times 10^{-16}$, so neither solver loses accuracy as n grows. QR's mean residual is about twice LU's, consistent with its roughly doubled operation count.

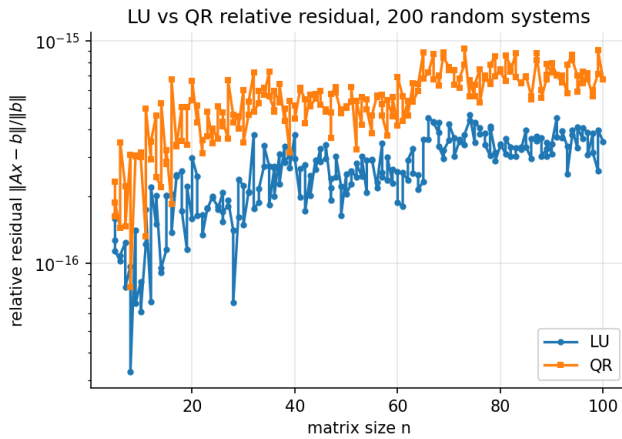
Table 3 shows the Hilbert stress test. Here the residual stays tiny but the error in x grows with the condition number, following

Table 1: Benchmark design for the numerical building blocks and the scripts that regenerate them.

Component	Test design	Independent truth	Script
LU, QR	200 random systems, $n = 5$ to 100; Hilbert $n = 6, 8, 10, 12$	manufactured x ; NumPy	m1_linalg_benchmark.py
Trapezoid, Simpson	$\int_0^\pi \sin x$, $\int_0^1 e^x$, $\int_0^1 (1+x^2)^{-1}$; $n = 9$ to 2001 points	exact values 2, $e - 1$, $\pi/4$	m1_quadrature_benchmark.py
RNG	10^6 uniforms and normals (seed 123); 2×10^5 Poisson draws per λ	theoretical moments, χ^2 , Φ	m1_rng_benchmark.py
Forward model	all 4 regions, all 25 frames; ϕ_1 sweep; grid refinement; Patlak fit	Radau and RK45 ODE solvers	m2_forward_model.py
Jacobian, condition- ing	central differences, relative step 1.5×10^{-5} ; full spectrum	finite differences; NumPy eigenvalues	m3_jacobian_and_solver.py
IRGNM recovery	20 seeds per δ_x , 300 iterations, noiseless data	hidden ground truth	m3_irgnm_recovery.py

Table 2: Verification summary for the hand-written numerical stack.

Component	Measured result
LU vs NumPy, 200 systems	max relative error 6.48×10^{-16}
QR vs NumPy, 200 systems	max relative error 1.13×10^{-15}
Trapezoid order	1.9999–2.0008
Simpson order	3.9987 and 4.0125; 5.9988 for a cancellation case
Closed form vs Radau	1.73×10^{-13}
Quadrature vs Radau	3.29×10^{-10}
Jacobian vs finite differences	3.62×10^{-6} (λ), 1.72×10^{-6} (μ), 1.27×10^{-7} (m), 2.85×10^{-8} (kinetics)
Power method, largest eigen- value	22320.478409382842 vs 22320.478409382828

**Figure 5: Relative residual of our LU and QR solvers on 200 random systems of size 5 to 100. Both stay below 10^{-15} at every size.**

the rule of thumb that about $\log_{10} \kappa(A)$ of the 16 available decimal digits are lost. At $n = 12$, $\kappa \approx 1.8 \times 10^{16}$ leaves at most one or two correct digits, and LU, QR and NumPy all fail, by different amounts. This is a property of the matrix, not of any implementation, and it is the same mechanism that makes the parameter-identification problem need regularization.

Table 3: Hilbert systems with exact solution $x = (1, \dots, 1)$: relative error of the computed solution.

n	$\kappa_2(H)$	LU	QR	NumPy
6	1.50×10^7	2.54×10^{-10}	4.25×10^{-11}	1.42×10^{-10}
8	1.53×10^{10}	1.20×10^{-7}	1.93×10^{-7}	6.12×10^{-8}
10	1.60×10^{13}	2.68×10^{-4}	2.13×10^{-4}	8.67×10^{-5}
12	1.76×10^{16}	3.91×10^{-2}	1.73×10^{-1}	3.25×10^{-1}

Table 4: Error of $\int_0^\pi \sin x dx$ as the grid is refined. The ratio is the error at the previous n divided by the error at this n .

n	trapezoid error	ratio	Simpson error	ratio
9	2.58×10^{-2}	—	2.69×10^{-4}	—
17	6.43×10^{-3}	4.01	1.66×10^{-5}	16.23
33	1.61×10^{-3}	4.00	1.03×10^{-6}	16.06
65	4.02×10^{-4}	4.00	6.45×10^{-8}	16.01
129	1.00×10^{-4}	4.00	4.03×10^{-9}	16.00
257	2.51×10^{-5}	4.00	2.52×10^{-10}	16.00
513	6.27×10^{-6}	4.00	1.57×10^{-11}	16.00
1025	1.57×10^{-6}	4.00	9.85×10^{-13}	15.98

4.1.2 Integration. Halving the step must divide the trapezoid error by $2^2 = 4$ and the Simpson error by $2^4 = 16$. Table 4 shows exactly this for $\int_0^\pi \sin x dx = 2$; the fitted orders are 2.0008 and 4.0125. For e^x on $[0, 1]$ they are 1.9999 and 3.9987. For $f(x) = 1/(1+x^2)$ on $[0, 1]$, Simpson measures order 5.9988: here $f'''(x) = 24x(1-x^2)/(1+x^2)^4$ vanishes at both endpoints, so the h^4 term of (7) cancels and the h^6 term is what we see. With local-coordinate weights the Simpson error keeps falling to 6.6×10^{-14} at $n = 2001$ for $\sin x$ and reaches the float64 limit (2.2×10^{-16}) for the cancellation case (Figure 6).

4.1.3 Random numbers. Table 5 summarizes the generator diagnostics. The uniform stream has the theoretical mean and variance to four digits. A chi-square test over 10 equal bins of 10^6 draws gives 12.68, below the 5% critical value 16.92 for 9 degrees of freedom, so uniformity is not rejected. The Box–Muller normals have mean, skewness and excess kurtosis near 0 and variance near 1, and their empirical distribution function stays within 5.6×10^{-4} of Φ on $[-3, 3]$. Lag-1 to lag-5 autocorrelations of the uniform stream all lie inside one standard error ($1/\sqrt{n} = 10^{-3}$) of zero. That matters because Box–Muller is known to expose correlations between

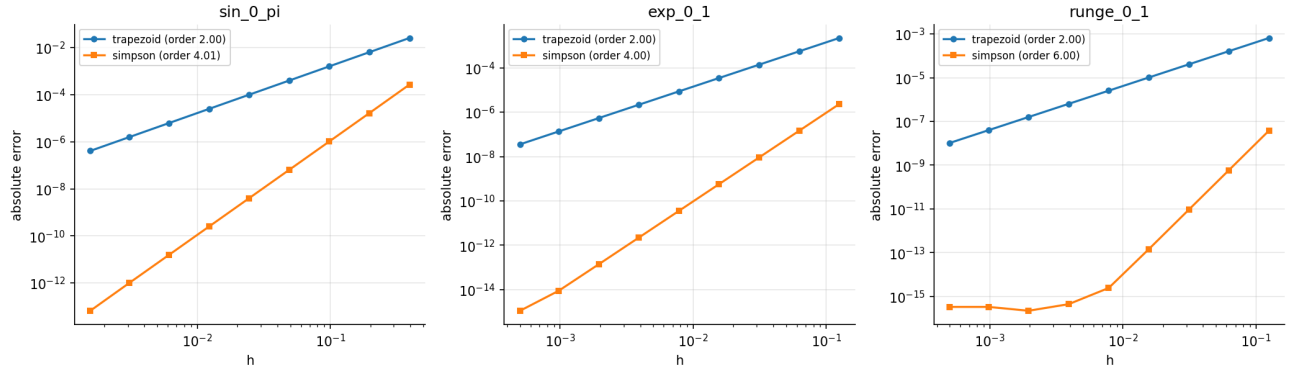


Figure 6: Absolute error against step size h on log-log axes; the slope is the convergence order. Trapezoid has slope 2 and Simpson slope 4, except for $(1 + x^2)^{-1}$, where the endpoint cancellation in (7) gives slope 6 until the error reaches machine precision.

Table 5: Random-number diagnostics ($n = 10^6$ unless stated).

Statistic	Measured	Target
Uniform mean / variance	0.499721 / 0.083412	0.5 / 0.083333
χ^2 , 10 bins, 9 d.o.f.	12.68	< 16.92 (5%)
Normal mean / variance	0.000388 / 1.000037	0 / 1
Normal skew / excess kurtosis	0.0035 / 0.0038	0 / 0
$\max \hat{F} - \Phi $ on $[-3, 3]$	5.6×10^{-4}	$\rightarrow 0$
Autocorrelation, lags 1–5	$ \rho \leq 4.0 \times 10^{-4}$	$ \rho < 10^{-3}$
Poisson $\lambda = 4$ (Knuth)	4.0013 / 4.0046	4 / 4
Poisson $\lambda = 25$ (Knuth)	24.991 / 25.012	25 / 25
Poisson $\lambda = 1000$ (normal)	1000.004 / 1002.29	1000 / 1000

Table 6: Maximum relative difference over the 25 frames between the closed form and independent evaluations of C_T .

Region	vs Radau	vs RK45	vs Simpson
Frontal	1.12×10^{-13}	6.93×10^{-12}	3.18×10^{-10}
Temporal	1.73×10^{-13}	8.29×10^{-12}	3.29×10^{-10}
Occipital	9.54×10^{-14}	6.03×10^{-12}	2.40×10^{-10}
White matter	1.37×10^{-13}	1.42×10^{-11}	2.47×10^{-10}

consecutive LCG outputs (the Neave effect). Both Poisson branches reproduce mean = variance = λ within Monte Carlo error. Figure 7 shows the histograms and the consecutive-pair scatters; the uniform pairs fill the square without the lattice stripes of a poor LCG.

4.1.4 Forward model. Table 6 breaks the three-way agreement down by region. The closed form agrees with the stiff Radau ODE integration to about 10^{-13} , with the explicit RK45 integration to about 10^{-11} , and with our graded-grid Simpson path to about 3×10^{-10} . Because the three paths share no algebra, an error in the closed-form derivation would have shown up as a disagreement.

Table 7 shows why ϕ_1 needs care. The literal formula $(e^x - 1)/x$ loses about half its digits by $x = 10^{-8}$, is wrong by 11% at $x = 10^{-15}$, and returns 0 instead of 1 (relative error 1) from $x = 10^{-16}$ down, while the expm1 form is exact to machine precision everywhere. The near-degeneracy sweep in Figure 8 tests the related case $a + \mu \rightarrow 0$

Table 7: Relative error of $\phi_1(x)$ evaluated literally and with expm1.

x	$(e^x - 1)/x$	expm1(x)/ x
10^{-4}	4.3×10^{-13}	0
10^{-8}	1.1×10^{-8}	0
10^{-12}	8.9×10^{-5}	0
10^{-15}	1.1×10^{-1}	0
10^{-16}	1.0	0

Table 8: Relative error of the Simpson path at the last frame ($t = 57.5$ min) for grids $t_i = t_{\max} u_i^q$.

n	$q = 1$ (uniform)	$q = 2$	$q = 3$ (adopted)
401	1.17×10^{-3}	1.24×10^{-7}	8.12×10^{-8}
801	9.68×10^{-5}	7.77×10^{-9}	5.08×10^{-9}
1601	6.56×10^{-6}	4.86×10^{-10}	3.18×10^{-10}
3201	4.18×10^{-7}	3.04×10^{-11}	1.99×10^{-11}
12801	1.64×10^{-9}	1.18×10^{-13}	7.70×10^{-14}

in (11): across the singular point, the closed form and the quadrature agree to 3.06×10^{-10} , with no spike.

The grid for the quadrature path was chosen by measurement (Table 8). The hardest frame is the last one (57.5 min), because the integral accumulates the most error by then. With a uniform grid ($q = 1$), 1601 points leave a relative error of 6.6×10^{-6} there; the cubic grading $q = 3$ reduces it to 3.2×10^{-10} at the same cost, and at this frame $q = 3$ is the best of the gradings $q = 1, \dots, 5$ at every n we tested.

As an end-to-end use of the QR least-squares routine, we fitted Patlak lines [11] to frames 13–25 ($t \geq 3.5$ min), plotting C_T/C_P against $\int_0^t C_P ds/C_P$ (Figure 9, Table 9). The fitted slopes match the classical net influx rate $K_i = K_1 k_3 / (k_2 + k_3)$ to 0.1–1.6%, with $R^2 \geq 0.997$. A closer look shows that the classical K_i is itself only approximate for this input. Because the slowest input exponent $\mu_4 = -0.0106$ is not negligible, the late-time uptake rate $\dot{C}_T(t)/C_P(t)$ tends to $K_1(k_3 + \mu_4)/(k_2 + k_3 + \mu_4)$ instead of K_i . At the last frame,

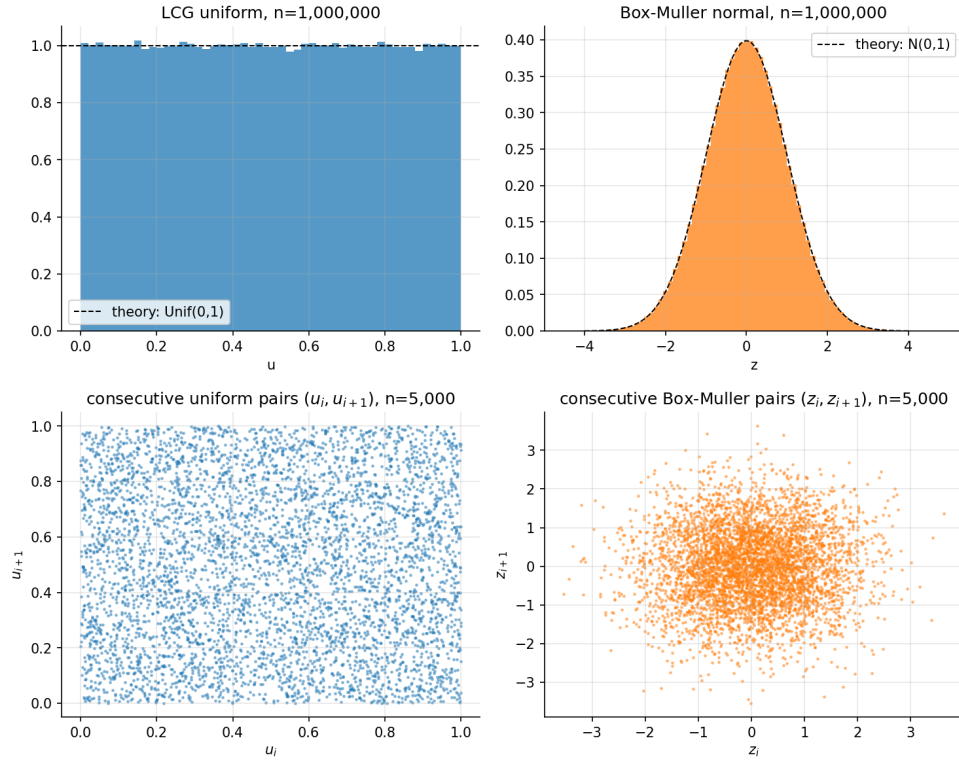


Figure 7: Top: histograms of 10^6 LCG uniforms and of 10^6 Box–Muller normals with the $N(0, 1)$ density overlaid. Bottom: 5000 consecutive pairs (u_i, u_{i+1}) , which fill the unit square uniformly, and 5000 consecutive normal pairs (z_i, z_{i+1}) , which form a round cloud with no visible structure.

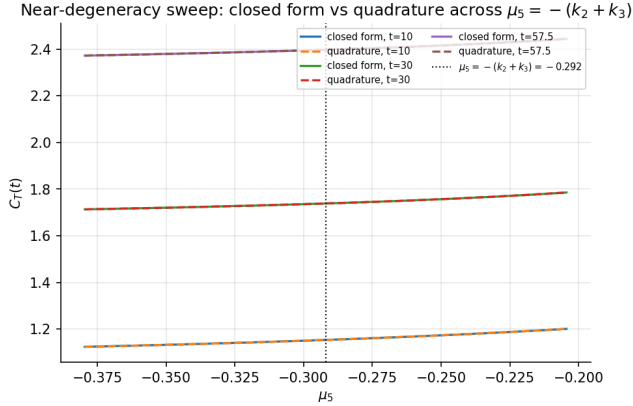


Figure 8: Closed form and quadrature for C_T at $t = 10, 30$ and 57.5 min while one input exponent μ_5 is swept through $-(k_2 + k_3)$, where the naive kernel is $0/0$. The curves coincide and stay smooth across the dotted line.

the classical value is off by 5.6–18.7%, whereas this corrected value is within 0.05–0.34%, and at $t = 800$ min it is matched to better than 8×10^{-16} .

Table 9: Patlak slopes fitted with our QR least squares, and the late-time uptake rate $\hat{C}_T/C_P$ at $t = 57.5$ min compared with the classical and corrected formulas.

Region	K_i	slope	R^2	class. err.	corr. err.
Frontal	0.0634	0.0633	0.9999	5.6%	0.05%
Temporal	0.0562	0.0561	0.9998	7.5%	0.08%
Occipital	0.0631	0.0631	0.9996	8.2%	0.10%
White matter	0.0226	0.0229	0.9970	18.7%	0.34%

4.1.5 Jacobian, conditioning and noiseless recovery. The full Jacobian has condition number 7.316×10^4 . Its normal equations have condition number 5.353×10^9 , illustrating the expected squaring. Along the regularization schedule, the normal-equation condition number grows from 3.38×10^2 to 1.43×10^{12} , whereas the stacked QR matrix grows from 18.4 to 1.20×10^6 (Figure 10). The two paths compute the same step to 2×10^{-16} at the start, but the gap widens to 5.0×10^{-10} by the end of the schedule, as the squared condition number predicts. Consequently QR is the production solver.

On the current whole-problem Track A/B check (noiseless tissue and blood data, $\delta_x = 0.1$, seed 0), regularized IRGNM reaches relative parameter error 9.27×10^{-7} after 300 iterations in 0.599 s. SciPy’s unregularized trust-region least-squares method, given the same residual, analytic Jacobian, and box bounds, terminates after

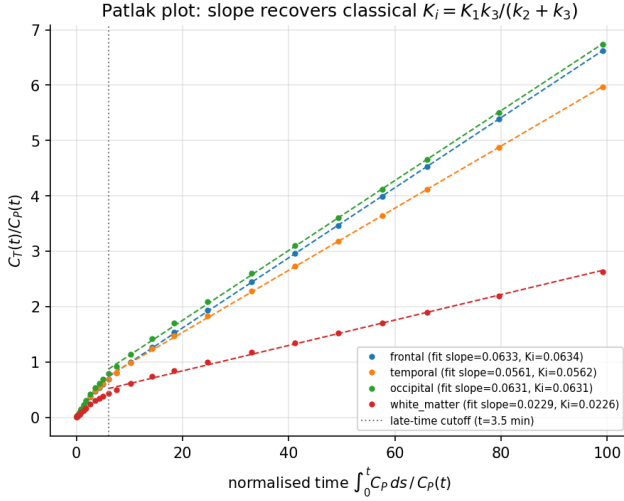


Figure 9: Patlak plot for the four regions. Late points (right of the dotted line, $t \geq 3.5$ min) fall on straight lines whose QR-fitted slopes recover K_i .

Table 10: Noiseless IRGNM recovery, 20 seeds per perturbation level, 300 iterations. “Clipped” counts starts that had to be projected into the feasible box.

δ_x	diverged	clipped	median error	mean iter.
0.1	0/20	0	4.26×10^{-7}	300.0
0.2	3/20	0	5.21×10^{-7}	276.1
0.3	5/20	2	5.41×10^{-7}	267.6
0.4	8/20	3	9.80×10^{-7}	249.3

43 function evaluations in 0.056 s at error 0.094: better than its starting error of 0.301, but about 10^5 times less accurate than IRGNM. This is not a claim that SciPy is inaccurate: the two optimizers solve differently regularized problems, and the comparison shows why the paper’s Tikhonov schedule matters for this ill-posed inverse problem.

The tissue-only Jacobian annihilates the predicted scale direction with norm 9.59×10^{-17} compared with matrix scale 166.1, and inverse iteration aligns with that direction with cosine 1. Adding the blood block raises the smallest eigenvalue to 4.17×10^{-6} . Figure 11 shows the full spectrum.

Noiseless IRGNM recovery from 20 starts per perturbation level produced median relative errors 4.26×10^{-7} , 5.21×10^{-7} , 5.41×10^{-7} , and 9.80×10^{-7} for $\delta_x = 0.1, 0.2, 0.3, 0.4$. Divergence counts were 0, 3, 5, and 8 of 20, respectively; no failed trajectory was reseeded away (Table 10). Figure 12 shows both the characteristic late error decrease and the increasing sensitivity to initialization. In the representative run ($\delta_x = 0.1$, seed 0), the data misfit falls by a factor of about 45 in the first 100 iterations while the parameter error barely changes: the solver first fixes the directions the data constrain strongly, and only later the nearly flat directions identified by the conditioning analysis.

4.2 Identifiability signature

Table 11 is the central result. Across 65 accepted tissue-only fits, the common-ratio spread never exceeds 1.83×10^{-8} even though ζ ranges from 0.693 to 2.269 and the largest K_1 relative error is 127%. In the same fits, the maximum k_2 and k_3 errors are only 1.33×10^{-6} and 5.11×10^{-7} . This separation is direct numerical evidence for the paper’s structural theorem, rather than merely a successful curve fit.

The best single sample occurs at $t = 0.2917$ min, where $C_P = 4.7995$; late samples at 15 and 57.5 min are 20–60 times less effective. Under the full-step RMS convention, tissue-only median spreads are about 0.49–0.54% at high count and 1.76–1.96% at normal count, while ζ still varies substantially. At low count only 2 of 80 tissue-only fits are accepted, so the signature is no longer distinguishable. Adding one early C_P sample yields median $|\zeta - 1|$ of 0.66–0.79% at high count and 2.47–3.02% at normal count, but no accepted low-count fits. Correct Morozov early stopping improves estimation stability but masks the asymptotic ratio coincidence because it stops before the iterate fully settles onto the null manifold.

4.3 Noise, consistency, and regularization

The calibrated Poisson noise produced mean RMS discrepancies 0.002958 ± 0.000261 , 0.011616 ± 0.001076 , and 0.070341 ± 0.006409 for high, normal, and low count. In the controlled consistency experiment with one exact early plasma sample and $\delta_x = 0.1$, mean kinetic error decreases from 0.14430 ± 0.04052 at normal count to 0.12109 ± 0.04940 at high count and to $(1.97 \pm 1.04) \times 10^{-7}$ without noise. All 20 low-count cases stop at iteration zero, correctly indicating that the data do not justify moving away from the initial guess.

Regularization reduces the sum of truth-normalized fitted-parameter variances from 0.28668 to 0.16263 at high count (factor 1.76) and from 1.35754 to 0.30952 at normal count (factor 4.39), measured on 14 paired successful seeds in each level. A fixed-Jacobian one-step diagnostic, which retains all 20 paired observations, gives still larger variance-reduction factors of 2231, 1251, and 420 for high, normal, and low count. Figure 13 summarizes both checks.

4.4 Monte Carlo trends and comparison with the paper

The original 960-cell grid records 494 non-finite divergences: 28 noiseless, 76 high-count, 153 normal-count, and 237 low-count. Thus failure probability grows sharply with noise; Figure 14 shows typical noisy trajectories. Among finite M4.3 runs, mean per-seed K_1 versus k_3 relative errors are 0.0106 versus 0.1576 in Setup B and 0.0160 versus 0.0850 in Setup C, agreeing with the paper that K_1 is generally easier to recover. Setup A reverses the ordering (0.2307 versus 0.1737), as expected because it deliberately removes arterial data and exposes the K_1/λ ambiguity.

For $\delta_x = 0.1$, known C_P beats clean C_{WB} at both high count (0.1211 vs 0.1324 kinetic error) and normal count (0.1443 vs 0.4149). Clean C_{WB} beats noisy C_{WB} at high count (0.1324 vs 0.1668) but reverses at normal count (0.4149 vs 0.3582), where only 7 runs per cell survive. We therefore classify that comparison as mixed rather than smoothing over the small-sample result.

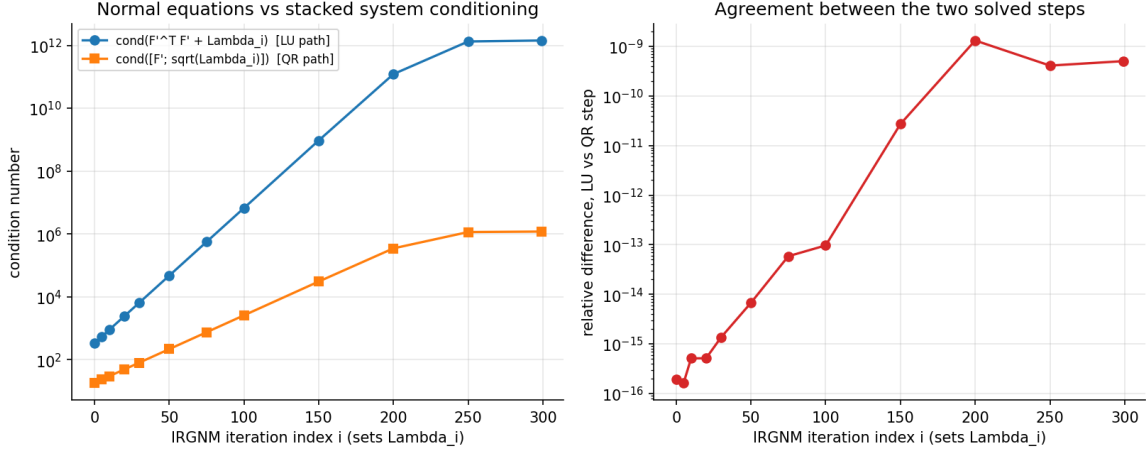


Figure 10: Left: condition number of the normal-equations matrix (LU path) and of the stacked matrix (QR path) along the IRGNM schedule; the first is roughly the square of the second. Right: relative difference between the steps computed by the two paths.

Table 11: Noiseless tissue-only identifiability and removal by one early C_P measurement. The spread is $\max_i r_i / \min_i r_i - 1$, $r_i = K_{1,\text{est}}^i / K_{1,\text{true}}^i$.

δ_x	Accepted/20	ζ range	Worst spread	Max K_1 error	Max k_2 error	Max k_3 error	One C_P : max $ \zeta - 1 $
0.1	18	0.838–1.335	7.59×10^{-9}	0.335	5.68×10^{-7}	2.19×10^{-7}	7.65×10^{-8}
0.2	17	0.697–1.472	1.83×10^{-8}	0.472	1.33×10^{-6}	5.03×10^{-7}	1.37×10^{-7}
0.3	17	0.693–1.862	1.77×10^{-8}	0.862	1.32×10^{-6}	5.11×10^{-7}	1.73×10^{-7}
0.4	13	0.754–2.269	1.52×10^{-8}	1.269	1.15×10^{-6}	4.45×10^{-7}	1.37×10^{-7}

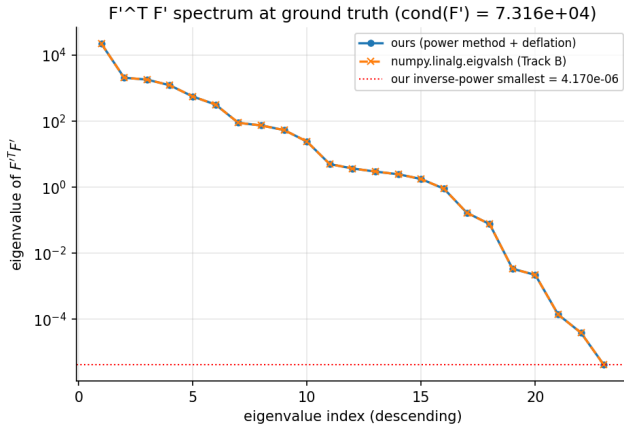


Figure 11: Spectrum of $F'^T F'$ at ground truth. Our power/deflation results overlay NumPy’s reference; inverse power iteration supplies the smallest eigenvalue.

The original grid uses a stopping rule with a scaling error and a single set of solver settings for all setups. Section 4.5 reruns it with each correction in turn: with the paper’s own settings, failure counts match the paper at high and normal count, and the remaining mismatch is concentrated in low count and noiseless distant starts.

Table 12: Predicted trends and measured verdicts.

Trend	Verdict and evidence
K_1 easier than k_3	Agrees in B/C; A reverses as Proposition 12 predicts
Known C_P better than clean C_{WB}	Agrees at high and normal count
Clean C_{WB} better than noisy C_{WB}	High agrees (14 and 13 survivors); normal reverses (7 each)
Low count fails	Agrees: 237/240 grid fits diverge
Regularization lowers variance	Agrees: factors 1.76 and 4.39 in full fits
Few failures (paper’s settings)	Agrees at high and normal count (14 vs 19, 18 vs 35); differs at low count and without noise

At low count, direct TAC noise often makes the discrepancy bound larger than the starting residual, producing a valid iteration-zero stop rather than a solver blow-up. These results support qualitative, not digit-for-digit, agreement; Table 12 lists each predicted trend with its verdict.

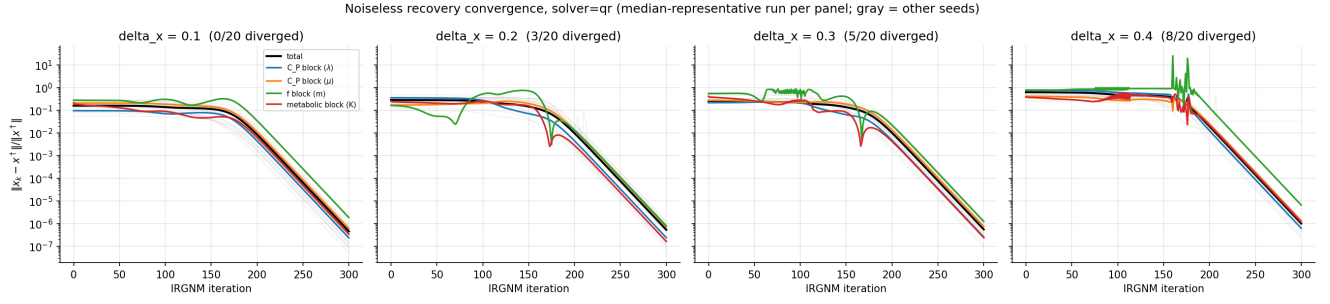


Figure 12: Noiseless IRGNM convergence. Colored curves are blockwise errors for a representative successful run; gray curves show other seeds.

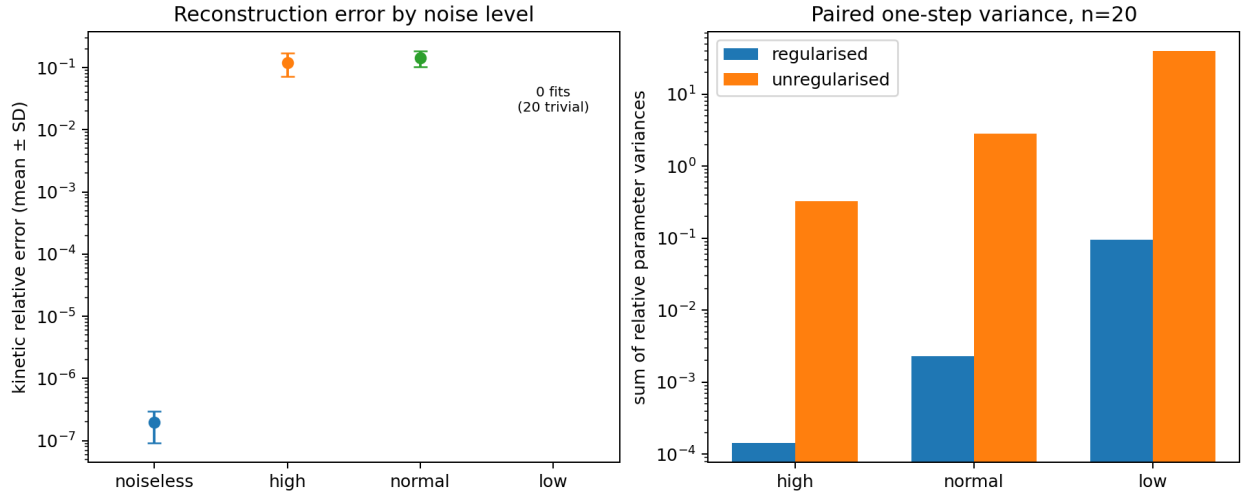


Figure 13: Left: kinetic reconstruction error decreases with the noise level; low count yields only trivial stops. Right: paired one-step parameter variance with and without regularization.

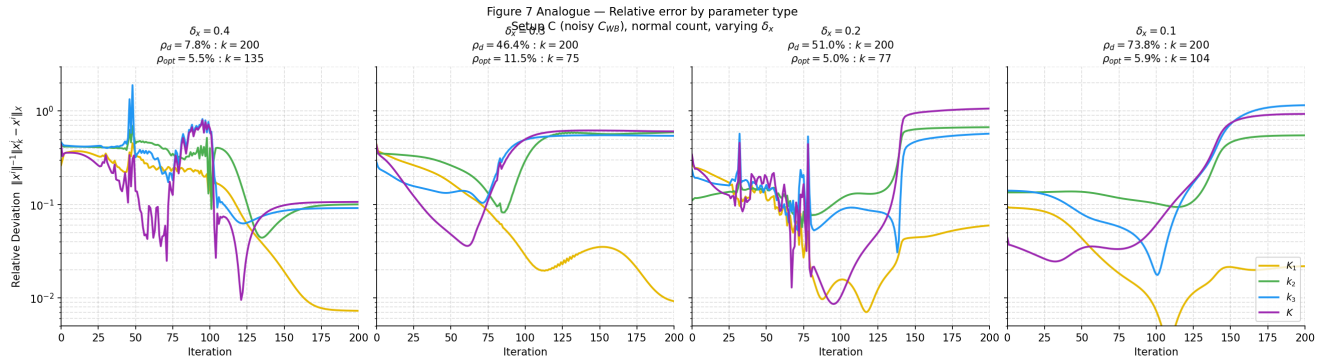


Figure 14: Current Figure 7 analogue for Setup C at normal count. Each panel shows the relative deviation of K_1 , K_2 , K_3 , and the combined kinetic block for one initialization level. The printed iteration and residual annotations expose early improvement followed by noise fitting or instability.

4.5 Effect of the improvements

Each improvement was evaluated by rerunning the same seeded experiments with only that change switched on, so the before-and-after values in Table 13 share observations, initial guesses, and seeds.

The base paper’s published settings were used exactly; nothing was tuned to match its results.

The stopping-rule correction accounts for most of the change. The original code compared an RMS noise level with the 2-norm of the residual, which made the discrepancy principle $\sqrt{100} = 10$ times too strict, so noisy runs never stopped early and many diverged. With the paper’s settings (Table 14), failure counts at high and normal count match the paper (14 vs 19, 18 vs 35). The paper’s settings raise the “no improvement” total from 172 to 222 because its larger stopping constants ($\tau = 9.2$ and 17.6) stop more low-count runs at once: 158 of the 161 low-count failures stop at iteration zero, where $\tau\delta$ already exceeds the initial residual, and none diverges. These are a consequence of adding noise directly to the TACs, not solver breakdowns. The remaining unexplained gap is noiseless: 29 of 240 runs from the most distant starts fail, against 1 in the paper. The stability changes left IRGNM’s reference result unchanged but improved SciPy’s from 1.046 to 0.094, because both solvers now receive more accurate model values; IRGNM remains about 10^5 times more accurate.

4.6 Floating-point stability and performance

The local-coordinate Simpson formulation removes a coordinate-dependent arithmetic defect. Translating the same quadratic samples by 10^3 changes the legacy error from 8.88×10^{-16} to 6.36×10^{-7} , while the local formula remains exact at float64 precision. On all PET regions and frames, increasing the graded grid to 12,801 points reduces maximum error to 7.99×10^{-14} with local weights, whereas the legacy formulation rises to 1.27×10^{-7} . Figure 15 shows that algebraically equivalent formulas need not be numerically equivalent.

Operation-count expectations and measured timing tell different stories at small sizes. LU and square QR are cubic in theory, but fitted exponents over $n = 20$ – 100 are 1.22 and 1.16 because interpreter and dispatch overhead dominate. The IRGNM stacked least-squares solve, measured over 23–736 columns, has exponent 2.60, closer to the cubic count. Simpson is linear and measures exponent 1.02 over 101–25,601 points. At the fixed physical problem size, the recorded IRGNM run averages 1.96 ms per iteration. Figure 16 shows the fits. Track B is faster because it uses optimized compiled kernels; its purpose here is validation, not replacement of the required hand-written algorithms.

4.7 Enhancements and design iterations

Several choices in the final pipeline replaced a simpler first version after a measurement showed a problem. Table 15 lists them with the evidence that motivated each one, including one enhancement that was implemented but not yet evaluated. Two of them deserve comment. First, our first Simpson implementation integrated each quadratic through differences of cubic antiderivatives in absolute coordinates. Its error stopped improving at $n = 513$ (2.10×10^{-10} for $\sin x$) and then grew to 4.81×10^{-9} at $n = 2001$. Rewriting the same weights in local coordinates (6) removed this floor: the error now keeps falling to 6.62×10^{-14} . Second, the Armijo backtracking line search accepts a step only if it decreases the regularized objective sufficiently, and falls back to a scaled projected-gradient direction when projection destroys descent. It targets the divergences from

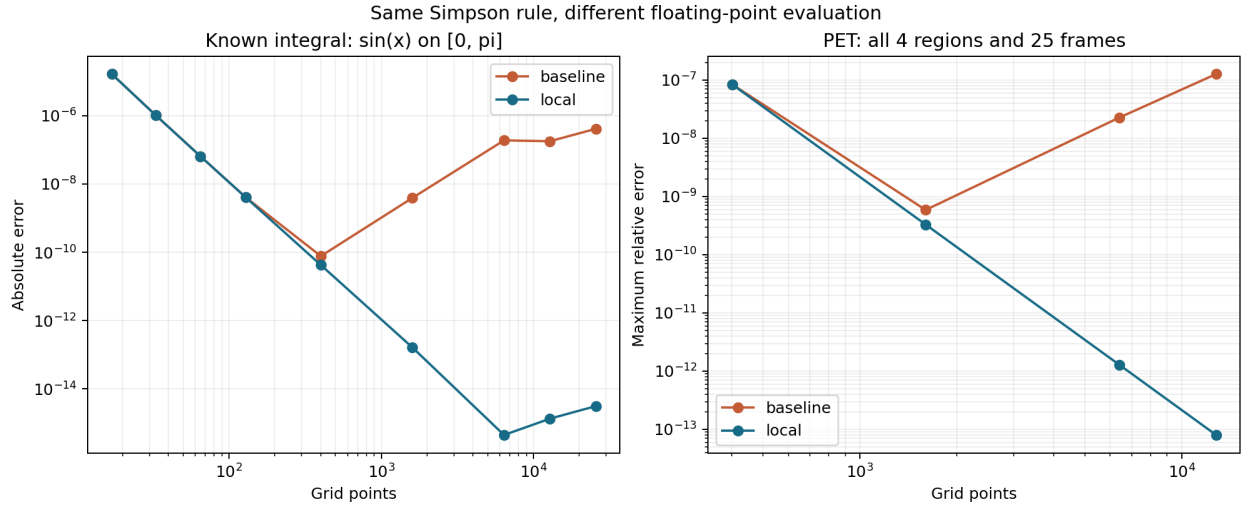
distant starts in Table 10. It is merged into the solver as an opt-in flag, off by default. On the reference noiseless fit it accepts the full step every time and reaches the same 9.27×10^{-7} , but it has not been run through the recovery study or the Monte Carlo grid, so we make no claim about its effect on divergence.

4.8 Course-topic mapping

Table 16 maps the relevant CSE 402 topics to concrete code and evidence. Methods not required by this inverse problem, such as Bairstow, golden-section search, Romberg integration, and interpolation of tabulated solutions, were not inserted artificially.

Table 13: Effect of each improvement on the measured results. Monte Carlo rows use the full 960-fit grid; “no improvement” is the paper’s failure criterion (final error not below the initial error).

Improvement	Measured quantity	Before	After
Corrected stopping-rule scale, 200-step cap	Monte Carlo blow-ups (non-finite)	494	58
Paper’s per-setup settings and reduced setup	Monte Carlo “no improvement”	512	172
	Monte Carlo blow-ups	58	43
Blood readings at all 25 frames (not adopted)	Monte Carlo “no improvement” (paper: 97)	172	222
	Monte Carlo blow-ups / “no improvement”	43 / 222	84 / 209
Local-coordinate Simpson weights	Simpson error, $\int_0^\pi \sin x \, dx$, $n = 12801$	1.8×10^{-7}	1.3×10^{-15}
	PET forward model, max relative error, $n = 12801$	1.27×10^{-7}	7.99×10^{-14}
Stable exponential kernels	Quadrature vs Radau, $n = 1601$	5.88×10^{-10}	3.29×10^{-10}
	SciPy least-squares final error (reference fit)	1.046	0.094
Armijo line search (opt-in)	IRGNM final error (reference fit)	9.27×10^{-7}	9.27×10^{-7}
	IRGNM final error (reference fit)	9.27×10^{-7}	9.27×10^{-7}

**Figure 15: The same Simpson rule evaluated with legacy absolute-coordinate weights and stable local-coordinate weights. Local evaluation removes the fine-grid round-off growth.****Table 14: Failure counts against the paper’s Table 1 under its “no improvement” criterion, out of 240 fits per noise level.**

Noise	Paper	Original	Paper’s settings
Noiseless	1	28	29
High count	19	78	14
Normal count	35	167	18
Low count	42	239	161
Total	97	512	222

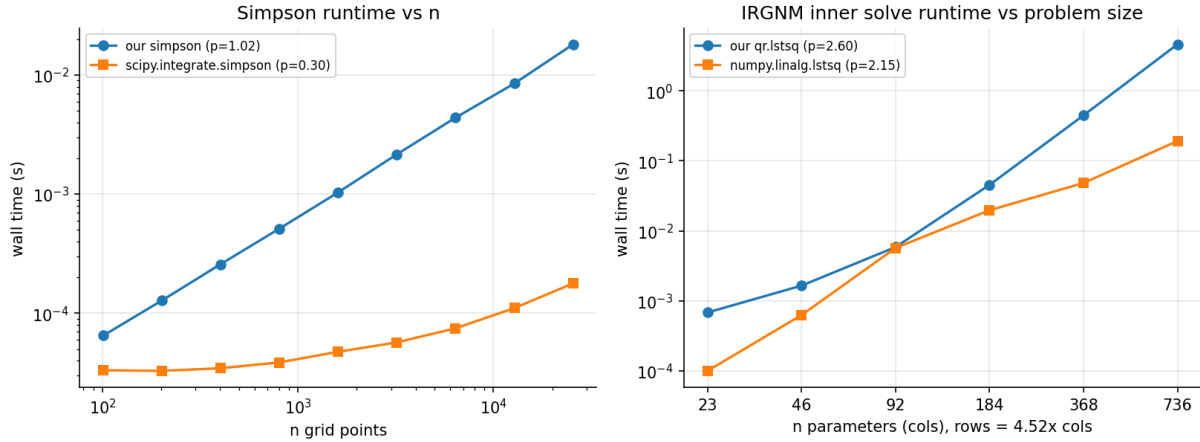


Figure 16: Measured scaling of Simpson integration and the stacked least-squares solve used inside IRGNM. Timing constants are machine-dependent; slopes are fitted by our QR least-squares implementation.

Table 15: Numerical enhancements tried during the project, the problem each addressed, and the measured outcome.

Enhancement	Problem addressed	Measured outcome	Status
Partial pivoting in LU	division by small pivots amplifies rounding	residual $\leq 4.7 \times 10^{-16}$ on 200 systems	adopted
Householder sign choice	cancellation in $x_1 - \alpha$	QR residual $\leq 9.2 \times 10^{-16}$	adopted
Stacked QR instead of normal equations	$\kappa(F'^T F' + \Lambda) \approx \kappa(\text{stacked})^2$	condition 1.20×10^6 instead of 1.43×10^{12}	adopted
Simpson on nonuniform grids	PET frames and graded grids are not uniform	orders 3.9987–4.0125 measured	adopted
Local-coordinate Simpson weights	cancellation in cubic antiderivatives	floor 2.10×10^{-10} (n=513) replaced by 6.62×10^{-14} (n=2001)	adopted
Graded grid $t = t_{\max} u^3$, $n = 1601$	early bolus needs dense sampling	last-frame error 3.18×10^{-10} vs 6.56×10^{-6} uniform	adopted
expm1-based ϕ_1	0/0 and cancellation at small x	relative error 0 vs 1.0 at $x \leq 10^{-16}$	adopted
Taylor series for ϕ_2 below 10^{-4}	cancellation persists in $e^x - 1 - x$	threshold measured; Jacobian agrees with finite differences	adopted
Factored kernel (11)	$0 \cdot \infty$ and $a + \mu \rightarrow 0$	smooth across the singular point, 3.06×10^{-10}	adopted
Poisson crossover at $\lambda = 30$	$O(\lambda)$ cost; endless loop for $\lambda \gtrsim 745$	mean and variance within Monte Carlo error up to $\lambda = 1000$	adopted
Seed mixing and derived seeds	related streams from nearby seeds	identical streams across processes	adopted
Corrected late-time slope	classical K_i ignores the slowest input exponent	error 5.6–18.7% reduced to 0.05–0.34%	analysis
Blockwise regularization	parameters span a scale ratio of 2690	noiseless recovery to $\approx 10^{-6}$	adopted
Corrected Morozov scale	RMS vs 2-norm mismatch in the stopping rule	non-finite failures 494 reduced to 58 (Table 13)	adopted
Paper’s per-setup settings	one setting used for all three setups	failures match the paper at high and normal count	adopted
Armijo line search	divergence from distant starts	same result on the reference fit; grid not yet evaluated	opt-in

Table 16: Course topics used in the project.

Topic	Implementation	Produced evidence
Round-off/truncation error	phi1, phi2 series, stable moments, local Simpson weights	Table 7; translated-quadratic and fine-grid study
Condition number and ill-conditioning	Hilbert stress test; Jacobian spectrum	Table 3: about $\log_{10} \kappa$ digits lost; $\kappa(F') = 7.3 \times 10^4$
Bisection root finding	calibrate_poisson, calibrate_gaussian	noise targets reached by log-scale bisection
LU/Gaussian elimination	src/linalg.py, Algorithm 1	6.48×10^{-16} max error over 200 systems
Householder QR	src/qr.py, Algorithm 2	stacked IRGNM solve and all log-log regressions
Power method	src/eigen.py	Jacobian conditioning spectrum and smallest eigenvalue
Random-number generation	src/rng.py	Table 5: moments, χ^2 , autocorrelation, Poisson
Monte Carlo simulation	src/montecarlo.py	960-cell seeded noise/initialization grid
Newton-type optimization	src/irgnm.py	regularized Gauss-Newton recovery and constraints
Least-squares regression	src/qr.lstsq; Patlak experiment [11]	Table 9: late-time Patlak $R^2 = 0.9970$ – 0.9999
Newton-Cotes integration	trapezoid, simpson	Table 4: second- and fourth-order convergence
Numerical differentiation	central-difference benchmark	independent analytic-Jacobian verification

5 Conclusion and Future Work

This project reproduces the base paper’s central identifiability phenomenon with an independently implemented numerical stack. The strongest result is not simply low fitting error: without arterial data, four independently parameterized regions return K_1 ratios agreeing to 1.83×10^{-8} while K_1 itself may be wrong by 127%, and one early plasma sample collapses the scale error to at most 1.73×10^{-7} . Correcting the stopping-rule scale and using the paper’s per-setup settings reduced Monte Carlo blow-ups from 494 to 43 and brought the high- and normal-count failure counts in line with the paper. Every primitive underneath that result was verified on its own: LU and QR at the round-off level, Simpson at its theoretical order, the random generators against their target distributions, and the forward model against two independent integrators. The Jacobian, conditioning analysis, and noiseless IRGNM recovery are also independently verified. The noisy experiments show both the benefit of regularization and the limits of low-count data.

The study has four main limitations. First, TAC-level noise is not a substitute for sinogram generation and OSEM reconstruction, so absolute divergence counts cannot be compared directly with the paper. Second, nonlinear convergence remains sensitive to distant initial guesses: 8 of 20 noiseless runs diverge at $\delta_x = 0.4$, 29 of 240 noiseless grid runs fail against 1 in the paper, and the default solver takes full steps; an Armijo-globalized option is implemented but not yet evaluated. Third, the normal approximation used for large-rate Poisson draws is efficient but approximate. Fourth, all data are synthetic and use one fixed four-region model.

Future work should evaluate the Armijo-globalized IRGNM that is already implemented, reproduce the scanner/reconstruction chain, evaluate exact or rejection-based large-rate Poisson samplers, optimize blood-sample timing under noise, and test the method on realistic or clinical TACs. Pinning the complete numerical environment and packaging the report build into the one-command runner would further strengthen reproducibility.

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